

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	29 June 2026
Team ID	SWTID-2026-5356
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority
Sprint-1	Dataset Collection	USN-1	As a developer, I want to collect the India Agricultural Crop Production dataset from reliable sources so that accurate analysis can be performed.	3	High
Sprint-1	Data Cleaning	USN-2	As a developer, I want to clean and preprocess the dataset so that missing/inconsistent values are removed.	5	High
Sprint-1	SQL Database	USN-3	As a developer, I want to create SQL tables and import datasets for analysis.	3	High
Sprint-1	SQL Queries	USN-4	As a developer, I want to write SQL queries to analyze EV charging stations, brands, prices, and range.	5	High
Sprint-2	Tableau Dashboard	USN-5	As a user, I want to view an interactive dashboard showing crop production insights across India.	8	High
Sprint-2	State-wise Analysis	USN-6	As a user, I want to compare crop production across different states to identify high-performing regions.	5	High

Sprint-2	Crop-wise Analysis	USN-6	As a user, I want to compare production of different crops so that I can identify the major crops in India.	3	Medium
Sprint-2	Seasonal Analysis	USN-8	As a user, I want to analyze crop production across different seasons to understand seasonal trends.	3	Medium
Sprint-3	Year-wise Analysis	USN-9	As a user, I want to analyse crop production from 1997–2021 to identify long-term trends.	5	High
Sprint-3	Efficiency Analysis	USN-10	As a user, I want to analyze crop yield efficiency by comparing crop production with cultivated area so that I can identify the most productive crops and regions.	5	High
Sprint-3	Tableau Story	USN-11	As a user, I want to explore the dashboard through an interactive Tableau Story for better understanding.	5	High
Sprint-4	Flask Website	USN-12	As a user, I want to access the Tableau Dashboard and Story through a Flask web application.	5	High
Sprint-4	Documentation	USN-13	As a developer, I want to prepare project documentation and screenshots for submission.	3	Medium
Sprint-4	GitHub Repository	USN-14	As a developer, I want to publish the complete project on GitHub for sharing and version control.	3	High
Sprint-4	Final Testing	USN-15	As a developer, I want to test all dashboards, web pages, and documentation before final submission.	4	High

Velocity:

The team achieved an average velocity of 20 story points per sprint (or 2 story points per day) based on a 10-day sprint duration.

$$Av = \frac{\text{Total Story Points Completed}}{\text{Number of Sprints}}$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.kaggle.com/datasets>

<https://data.gov.in>

<https://agriwelfare.gov.in>

<https://www.fao.org>

<https://help.tableau.com>